① Exercise 1

Bushelori Thulasizwe
Data Science

Select Statements

① Write a SQL query to retrieve all columns from employees table.

```
Select *
From brightlearn.databases.exercise_1.data;
```

Id	First Name	Last Name	Department	Salary	hire date	City
1	John	~~Davis~~ Doe	IT	55000	2018-06-15	New York
2	Jane	~~Wilson~~ Smith	HR	48000	2019-07-20	Chicago
3	Mike	~~Moore~~ Johnson	Finance	60000	2017-09-12	Los Angeles
4	Sarah	~~Clark~~ Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2018-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-10	Chicago
10	Laura	Hall	IT	50000	2020-08-16	San Francisco

Laura	Hall	50000	

Exercise 1 ②

Butlekai Thulasiram
Data Science.

② Write SQL query to find all the unique departments in Employees table

```
Select distinct department
From brightlearn databases.exercise-1-data;
```

Department
IT
HR
Finance
Marketing

③ Write a SQL query to retrieve all employee First and Last name, ordered by salary descending order

```
Select First_Name,
       Last_Name,
       Salary
From brightlearn databases.exercise-1-data
order by salary desc;
```

First Name	Last Name	Salary
Emily	Davis	62000
Mike	Johnson	6000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000

First name	Last name	Salary
Jane	Smith	48000

③ Exercise 1 Buthelezi Thulasiawe
Data Science

④ Write a SQL query to retrieve the top 5 highest-paid employees.

```
Select First_Name,
       Last_Name,
       Salary
From brightlearn.databases.exercise_1_data
Order by salary desc
Limit 5;
```

First_Name	Last_Name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000

⑤ Write a SQL query to find employees who work in the IT department

```
Select *
From brightlearn.databases.exercise_1_data
where department='IT';
```

id	First Name	Last Name	department	salary	hire date	city
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
10	Laura	Hall	IT	58000	2020-08-10	San Francisco

④ Exercise 1

Thulasizwe Butelezi
Data Science

⑥ Write a SQL query to find Employee who work in the Finance department AND have salary greater than 58 000 → Select * From brightlearn.exercise_1_data where department = 'Finance' and salary > 58000;

Id	First Name	Last Name	department	Salary	hiredate	City
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
7	Robert	Wilson	Finance	59000	2019-10-01	Houston

⑦ Write SQL query to find employees who work in the HR department or Marketing department

Id	First Name	Last Name	department	Salary	hiredate	City
2	Jan	Smith	HR	48000	2019-07-20	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisc
8	Jessica	Moore	HR	51000	2018-05-22	Los Angel
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

Select *
From brightlearn.databases.exercise_1_data
where department = 'HR' or department = 'Marketing';

⑧ Write a SQL query to find employees who do not work in IT department

Select *
From brightlearn.databases.exercise_1_data
where not department = 'HR';

Id	First Name	Last Name	department	Salary	hire-date	New York
1	John	Doe	IT	55000	2018-6	Los Angeles
3	Mike	Johnson	Finance	60000	2017-09	New York
4	Sarah	Brown	IT	59000	2021-03	San Francisc
5	David	White	Marketing	52000	2016-04	Chicago
6	Emily	Davis	IT	62000	2020-02	Houston
7	Robert	Wilson	Finance	59000	2019-10	Chicago
9	Daniel	Clark	Marketing	53000	2022-06	Chicago
10	Laura	Hall	IT	55000	2021-08	San Fran

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Exercise 1

Bottleneck Thularizer
Data Science

9 Write a SQL query to find employees who are in the HR, IT, or Finance department

```
Select *
From brightlearn.databases.exercise_1_data
Where department in ('HR', 'IT', 'Finance');
```

id	First_Name	Last_Name	department	Salary	hire_date	City
1	John	Doe	IT	55 000	2018-06-15	New York
2	Jane	Smith	HR	48 000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-04-30	Los Angeles
4	Sarah	Brown	IT	53 000	2021-03-25	New York
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
10	Laura	Hall	IT	50 000	2020-08-10	San Francisco

10 Write a SQL query to find employees who are in the IT department ~~[illegible]~~ have salary greater 50000 and are located in New York

```
Select *
From brightlearn.databases.exercise_1_data
Where department = 'IT'
and Salary > 50000
and City = 'New York';
```

id	First_Name	Last_Name	Department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-04-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

⑥ Exercise 2. Bootcamp Tutorial Data Science

(11) Write a SQL query to retrieve the first and lastname of employees who work in Finance ~~department~~ and Marketing department, earn more than 52 000 and order the results by salary in descending order.

```
Select First_Name,
       Last_Name,
       department,
       Salary
from brightlearn.database.exercise_1_data
where department in('Finance','Marketing')
and Salary > 52000
order by Salary desc;
```

First Name	Last Name	Department	Salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000
Daniel	Clark	Marketing	53 000

(12) Write a SQL query to find all unique cities where employees work, excluding those in the IT and HR departments

```
select distinct city
from brightlearn.database.exercise_1_data
where not department in('IT','HR');
```

City
Chicago
Los Angeles
San Francisco
Houston

(13) Write SQL query to retrieve employees who are NOT in the Finance department, have a salary greater than 50 000 and order the results by hire date in descending order

```
Select *
from brightlearn.databases.exercise_1_data
where not department in('Finance')
and Salary>50000
Order by hire_date asc;
```

Id	First Name	Lastname	department	Salary	hire date	city
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
5	David	White	Marketing	52 000	2018-01-10	San Francisco
8	Jessica	Moore	HR	5000	2021-05-22	Los Angeles
1	John	Doe	IT	55 000	2018-03-15	New York
4	Sarah	Brown	IT	53000	2022-09-01	New York
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

(7) Exercise 1 Rutuleesi Thulasiram
Data Science

(11) write a SQL query to find the first 3 employees who work in either Chicago or Los Angels and belong to the IT or Marketing department

```
Select *
From brightlearn.databases.exercise_data
where city in ('chicago', Los Angeles')
and department in('IT','Marketing')
Limit 3;
```

id	First name	Last name	department	Salary	Hiredate	City
6	Emily	DAVIS	IT	62000	2015-02-14	Chicago
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago